SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 1: 1. 8-Puzzle Problem

Aim: To solve the 8-puzzle using A* search.

Python Program

```
from heapq import heappush, heappop

goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

def heuristic(state):
    return sum(
        abs(i // 3 - (v - 1) // 3) +
        abs(i % 3 - (v - 1) % 3)
        for i, v in enumerate(state) if v != 0
    )

def solve(start):
    pq = [(heuristic(start), 0, start, [])]
    visited = set()

    while pq:
        _, cost, state, path = heappop(pq)
        if state == goal:
            return path + [state]
        if state in visited:
            continue
        visited.add(state)
        z = state.index(0)
        r, c = divmod(z, 3)

        for dr, dc in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
            nr, nc = r + dr, c + dc
            if 0 <= nr < 3 and 0 <= nc < 3:
                nz = nr * 3 + nc
                new = list(state)
                new[z], new[nz] = new[nz], new[z]
                new = tuple(new)
                if new not in visited:
                    heappush(pq, (cost + 1 + heuristic(new),
                                  cost + 1, new, path + [state]))
    return None

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
path = solve(start)
if path:
    for state in path:
        for i in range(0, 9, 3):
            print(state[i:i+3])
        print()
else:
    print("No solution")
```

Sample Output

```
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

Result: The program demonstrates the stated Artificial Intelligence problem.